

Use the situation below to answer questions 1 – 3.

A new Microsoft Surface computer costs \$1275, but depreciates 16.5% each year as new technology comes out.

1. Is this situation exponential growth or decay?

Exponential decay

2. Two students, Josh and Eve were debating how to represent a function to model the value, V , after x years. Both of their functions are below. Explain who you agree with and why.

Josh	Eve
$V(x) = 1275(0.835)^x$	$V(x) = 1275(1.165)^x$

I agree with Josh because to calculate decay, subtract the rate from 1.
 $1 - 0.165 = 0.835$.

3. Write the correct function that models the context using both forms.

$V(x) = a(b)^x$	$V(x) = a(1 \pm r)^x$
$V(x) = 1,275(0.835)^x$	$V(x) = 1,275(1 - 0.165)^x$

Use the situation below to answer questions 4 – 6.

A new school opened with 225 students in 2021 and plans to increase by 13.3% per year until they reach full capacity.

4. Is this situation exponential growth or decay?

Exponential growth

5. Write an equation that models the population of the school, P , after x years since the opening of the school in 2021.

$P(x) = 225(1 + 0.133)^x$ or $P(x) = 225(1.133)^x$

6. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 \pm r)^x$
$P(x) = 225(1.133)^x$	$P(x) = 225(1 + 0.133)^x$

7. A three-bedroom house in the suburbs of Jacksonville sold for \$320,628. If the housing prices are expected to increase 2.8% annually, write a function that models the cost of a house, C , x years after it was sold.

$$C(x) = 320,628(1 + 0.028)^x \quad \text{or} \quad C(x) = 320,628(1.028)^x$$

8. A small local company reported a profit of \$175,000 last year but is struggling due to the pandemic. They are reporting a decrease of 13.6% each year. Write a function that represents the profit, P , after t years.

$$P(t) = 175,000(1 - 0.136)^t \quad \text{or} \quad P(t) = 175,000(0.864)^t$$

9. As of 2019, the world population is 8.215 billion people and is expected to grow at a rate of 1.2% per year. Write an equation to model the population growth, where $P(y)$ is the population in billions of people and y is the time in years.

$$P(y) = 8.215(1 + 0.012)^y \quad \text{or} \quad P(y) = 8.215(1.012)^y$$

10. A culture of 22 bacteria grows at a rate of 150% each hour. Tyler wrote the equation $B(t) = 22(1.5)^t$ to represent the population of the bacteria after t hours. Describe and correct the error.

Tyler should add 1 to 1.5.

$$B(t) = 22(1 + 1.5)^t \quad \text{or} \quad B(t) = 22(2.5)^t$$